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Question 1 (1)

Question 1

Are Context-free languages closed under complement? Prove it or provide a counter example. (20 Marks)

solution: Context-free languages are not closed under complement

proof: we set $\Sigma = \{a, b, c\}$

We know that CFL is closed under union. If CFLs are closed under complement, then by Morgan's law, CFL would be also closed under intersection.

$$A \cap B = \overline{\overline{A} \cup \overline{B}}$$

As A and B would be CFLs, $\overline{A} \cup \overline{B}$ would be CFL.

Define that

$$L_1 = \{a^i b^j c^k \mid i \geq 0, j \geq 0\}, L_2 = \{a^i b^j c^k \mid i \geq 0, j \geq 0\}$$

(a)

L_1 is CFL as $\begin{cases} S \rightarrow AC \\ A \rightarrow aAb \mid \epsilon \\ C \rightarrow cC \mid \epsilon \end{cases}$ which is CFL

(b) L_2 is CFL as $\begin{cases} S \rightarrow aSIT \\ T \rightarrow bTcT \end{cases}$ which is CFL

$$L_1 \cap L_2 = \{a^n b^n j^n \mid n \geq 0\}$$

then we need to prove that $\{a^n b^n c^n\}$ is not CFL using pumping lemma

for any CFL L , has $p \geq 1$, let length $\geq p$ string can be write $S = uvwxy$

satisfy that $\begin{cases} |vwx| \leq p & \textcircled{1} \end{cases}$

$\begin{cases} |vx| \geq 1 & \textcircled{2} \end{cases}$

$i \geq 0$, has $uv^iwx^iy \in L$ $\textcircled{3}$.

suppose that $S = a^p b^p c^p \in L$ ($\geq p$)

by pumping Lemma. we divide it into 5 parts satisfy that $\textcircled{1} \textcircled{2}$.

after branching, we found that all situations has $i \neq 1$ make $uv^iwx^iy \notin L$. thus the situation can't prove that L is CFL
thus $L_1 \cap L_2$ is not CFL.

Now we have that L_1, L_2 is CFL,
but $L_1 \cap L_2$ is not CFL
which is contradict to it CFL is closed
under complement, it is also closed under intersection
thus CFL is not closed under complement

Question 1 (2)

Are Turing-co-recognizable languages closed under concatenation? Prove it or provide a counter example. (20 Marks)

Solution: Turing-co-recognizable languages are not closed under concatenation

① We have counter example:

Let

$$\text{NONHALT} = \{ \langle M, w \rangle \mid M \text{ doesn't halt on input } w \}$$

the NONHALT is co-turing-recognizable because its complement $\text{HALT} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$ is turing-recognizable

②:

Now using an alphabet containing a separator #

$$A = \{ \langle M \rangle \# \langle w \rangle \mid \langle M, w \rangle \in \text{NONHALT} \} \quad B = \Sigma^*$$

Then $A \in \text{co-RE}$ (it is just NONHALT in a standard encoded form) and B is decidable thus $B \in \text{co-RE}$.

Δ : Consider $A \cdot B = \{ xy \mid x \in A, y \in B \}$.

as $B = \Sigma^*$

$$A \cdot B = A \Sigma^*$$

the set of all strings have some prefix in A .
Equivalently: $s \in A \cdot B \iff \exists$ a split $s = xy$ with $x \in A$.

But $x \in A$ means $x = \langle M \rangle \# \langle w \rangle$ and $\langle M, w \rangle$ never halts, which is a $\forall$ -type (for all time steps, it hasn't halted) property. Together with outer existential split, membership in $A \cdot B$ has an $\exists \forall$ pattern, which lies beyond $\text{co-RE}(\text{II})$ thus $A \cdot B \notin \text{co-RE}$.

thus we have $A, B \in \text{co-RE}$ but $A \cdot B \notin \text{co-RE}$.

thus it is not closed under concatenation

Question 2

Is the language $A_{CFG} = \{\langle G \rangle \mid G \text{ is a CFG that only generates letter } a\}$ decidable? Prove your conclusion. (e.g., $S \rightarrow a|b$ is not valid and $S \rightarrow a|aa$ is not valid.)

(30 Marks Total)

Solution: $A_{CFG} = \{\langle G \rangle \mid G \text{ is CFG that only generates letter } a\}$ is decidable

Proof: We describe a Turing machine that decides A_{CFG}

From the example ($S \rightarrow a|aa$ is not valid) only generates letter a means: $L(G) = a^*$

Decider D for A_{CFG} : only input $\langle G \rangle$.

① test whether $a \in L(G)$.

CFG membership is decidable convert to CNF and run CYK, so we can decide whether G generates string a .

② Test whether G generates any string other than a .
Let $R = \Sigma^* \setminus \{a\}$ since $\{a\}$ is regular and regular language is closed under complement. R is regular.

construct CFG G' for.

$$L(G') = L(G) \cap R$$

which is possible because CFLs are closed under intersection with regular languages

Then decide whether $L(G') = \emptyset$. Emptiness for CFGs is decidable (check if the start symbol can derive some terminal string)

Accept if (i) $a \in L(G)$ and (ii) $L(G') \neq \emptyset$.

Then $L(G) = \{a\}$. Thus Acc_H is decidable.

Question 3

Question 3

Is the language $T = \{\langle M \rangle \mid M \text{ is a TM and } \forall w \in L(M), |w| \leq 50 \text{ and } M \text{ accepts } w \text{ within 50 steps}\}$ decidable? Prove your conclusion.

(30 Marks Total)

Solution: The language $T = \{\langle M \rangle \mid M \text{ is a TM and } \forall w \in L(M), |w| \leq 50 \text{ and } M \text{ accepts } w \text{ in 50 steps}\}$ is undecidable.

proof:

Let

$\overline{\text{HALT}} = \{ \langle M, w \rangle \mid M \text{ does not halt on input } w \}$.

we prove that $\overline{\text{HALT}} \leq_m T$

given an arbitrary pair $\langle M, w \rangle$, we effectively construct a turing machine M' as follows

Construction of M'

on input x the machine M' work as follows

1. if $x \neq 0^s$, reject immediately
2. if $x = 0^s$, Stimulate the execution of M on input w :

$\begin{cases} \text{if } M \text{ halts on } w, \text{ then accept } x \\ \text{if } M \text{ does not halt on } w \text{ then loop forever.} \end{cases}$

The computation of M' is clearly computable.

Case 1: M does not halt on w .

In this one, M' never accepts 0^s , and others input are immediately rejected. Hence

$$L(M') = \emptyset$$

Since the universal condition in the definition

of T holds vacuously for empty language,
we have:

$$\langle M' \rangle \in T$$

Case 2: M halts on w

The M' accepts the string $0^{|w|}$, but
 $|0^{|w|}| = |w| > 50$,

which violates the length requirement in the
definition of T , thus.

$$\langle M' \rangle \notin T.$$

thus $\langle M, w \rangle \in \overline{\text{HALT}} \Leftrightarrow \langle M' \rangle \in T$.

If T were decidable, $\overline{\text{HALT}}$ would also be
decidable, however. $\overline{\text{HALT}}$ is undecidable, otherwise
 HALT would be decidable by complement the
decider. This contradiction implies T is undecidable.

T is undecidable.